

Analysis III

Home Work 1

Student: 26-712-224

September 23, 2026

Exercise 1 (Analysis II Recap, 1+1+1 P.)

Let $I = [a; b]$ be a compact interval in $\mathbb{R}$.

- (a) Explain the following statement: “The inclusion

$$T(I, \mathbb{R}) \subseteq S(I, \mathbb{R})$$

is dense with respect to the supremum-norm.” In particular, define the spaces $T(I, \mathbb{R})$ and $S(I, \mathbb{R})$ and the supremum-norm. (See Chapters 6 and 7 in the script.)

- (b) Explain briefly how the above can be used to define the (Riemann) integral

$$\int_a^b f(x)dx$$

for any $f \in S(I, \mathbb{R})$.

- (c) Construct a sequence of functions in $T(I, \mathbb{R})$ converging uniformly to $f(x) = x$ on $I = [0; 1]$. (You do not have to prove the uniform convergence.)

ANSWER:

- (a) In the Analysis II notes:

- $T(I, \mathbb{R})$ is defined to be the **set of step functions** with domain I - a (perfect) interval of $\mathbb{R}$ - and whose co-domain is $\mathbb{R}$.
- $S(I, \mathbb{R})$ is defined as the set of **regulated functions** - ie functions, $f : I \rightarrow \mathbb{R}$, which have both right and left hand limits $\forall x \in I$ and these limits are finite.
- The supremum-norm is defined on all bounded function $f : I \rightarrow \mathbb{R}$ as given by: $\|f\|_\infty = \sup_{x \in I} \|f(x)\|$; where, in this case, I have chosen the absolute value norm on $\mathbb{R}$

The ”Inclusion” statement means that, using the supremum-norm, any ”regulated function” can be approximated to by step functions with any degree of accuracy.

- (b) As $T(I, \mathbb{R})$ is dense in $S(I, \mathbb{R})$ we know that given any $\epsilon > 0$, we can find a sequence $(f_k)_{k \in \mathbb{N}}$ of step function in $T(I, \mathbb{R})$ which converges using the supremum norm to $f \in S(I, \mathbb{R})$. That is:

$$\|f - f_k\|_\infty < \epsilon \quad \forall k > N \quad (N \in \mathbb{N})$$

Under such conditions we could then define the Riemann integral of $f \in S(I, \mathbb{R})$ by:

$$\int_a^b f = \lim_{k \rightarrow \infty} \int_a^b f_k \quad (\text{where } I = [a, b])$$

(c) Firstly, we divide $[0, 1]$ into k evenly spaced intervals: $[0 = \frac{0}{k}, \frac{1}{k}, \frac{2}{k}, \dots, \frac{k-1}{k}, 1 = \frac{k}{k}]$. Now let us create the sequence of step functions $(f_k)_{k \in \mathbb{N}}$, where each $f_k \in T([0, 1], \mathbb{R})$, that converge uniformly to $f(x) = x$ on $[0, 1]$:

$$f_k(x) = \frac{j}{k} \quad \text{when } x \in \left[\frac{j}{k}, \frac{j+1}{k} \right) \text{ and } f_k(1) = 1$$

Exercise 2 (Null sets, 3+4 P.)

In this exercise we show that two sets are Lebesgue null.

- (a) Let $N = \mathbb{R}^{n-1} \times \{0\} \subset \mathbb{R}^n$. Using a similar strategy as in Example 11.1.4 (3) in the script, show that N is a set of measure zero.
- (b) Denote by

$$S^1 = \{x \in \mathbb{R}^2 : |x| = 1\}$$

the unit circle in $\mathbb{R}^2$. For $m \in \mathbb{N}$, take $x_1, \dots, x_m$ to be m equidistantly placed points on S^1 . Using squares of equal sizes centered at these points and eventually sending $m \rightarrow +\infty$, show that S^1 is a set of measure zero.

Hint: How large must these squares be so that they cover the circle? You may use without proof that the distance between two points on S^1 is at most as large as the angle between them.

ANSWER:

- (a) $N = \mathbb{R}^{(n-1)} \times \{0\}$ and we now wish to show that this set in $\mathbb{R}^n$ has measure zero.

I will use a slightly different strategy than in Example 11.1.4 (3) - which I find simpler.

Given $\epsilon > 0$ consider the sequence of **open** and **bounded** intervals:

$$U_m := (-m, m) \times (-m, m) \times \dots \times (-m, m) \times \left(-\frac{\epsilon}{2^{m+1}(2m)^{(n-1)}}, \frac{\epsilon}{2^{m+1}(2m)^{(n-1)}}\right)$$

where there are $(n-1)$ terms of the form $(-m, m)$. Then it is clear that we have an open cover for N :

$$N \subset \bigcup_{m=1}^{\infty} U_m \tag{2.1}$$

We can easily compute the Lebesgue measure for each U_m as:

$$\lambda_n(U_m) = (2m)^{(n-1)} \frac{\epsilon}{2^{m+1}(2m)^{(n-1)}} = \frac{\epsilon}{2^m} \tag{2.2}$$

From (2.1) and (2.2) we conclude that:

$$\begin{aligned} \lambda_n(N) &\leq \sum_{m=1}^{\infty} \lambda_n(U_m) \\ &\leq \sum_{m=1}^{\infty} \frac{\epsilon}{2^m} \\ &\leq \epsilon \end{aligned}$$

As ϵ is arbitrary we can conclude that N is a set of measure zero (in $\mathbb{R}^n$).

(b) We will denote the set of points that comprise the boundary of the unit circle S^1 by ∂S .

Consider the diagram below showing the m points $p_1, p_2, \dots, p_m$ that are equally spaced on the circumference of the circle. The angle, extended from the centre of the circle, to each

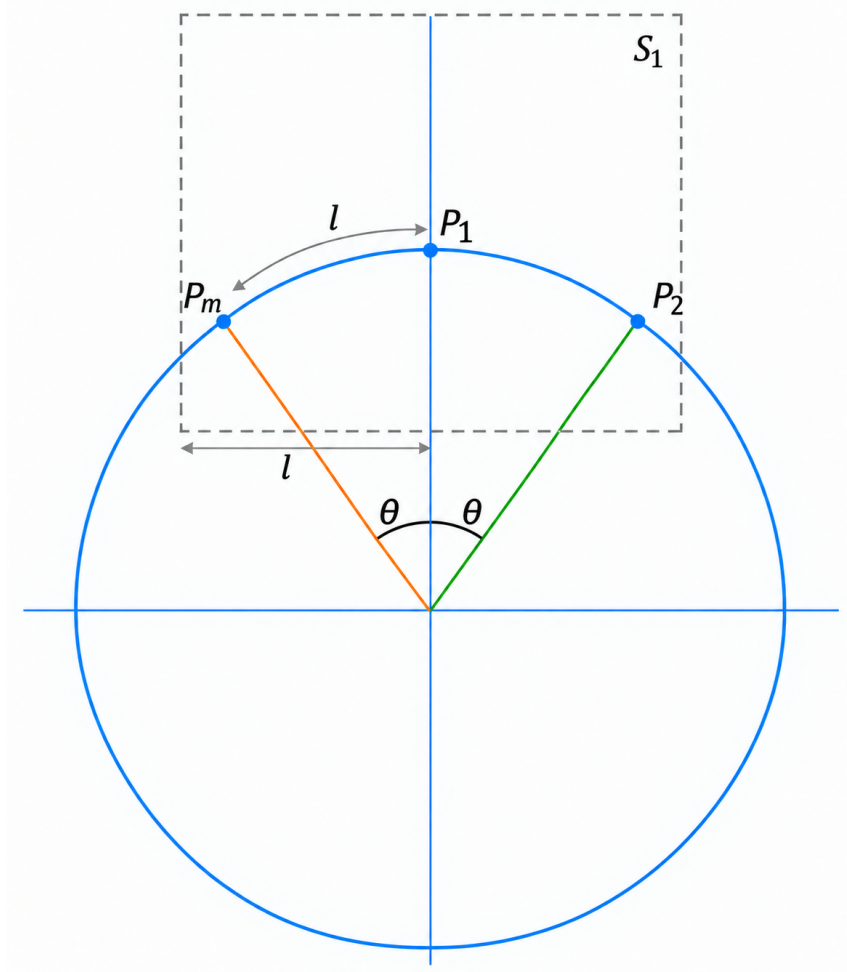


Figure 1: Circle construction

consecutive point on the circumference is $\theta = \frac{2\pi}{m}$. This means that, on the unit circle, the length of the arc between consecutive points is: $l = \frac{2\pi}{m}$.

Taking the point p_i as the centre of an **open** square of side length $2l = \frac{4\pi}{m}$, we can refer to this open square as S_i - An example of S_1 is shown in the diagram (but p_1 is a little "off centre" due to limitations of drawing).

It should be clear that the Lebesgue measure (i.e. area) of each such square is: $\lambda_2(S_i) = (2l)^2 = \frac{16\pi^2}{m^2}$.

The m squares $(S_i)_{i=1}^m$ - which are each **open** and **bounded** - overlap and completely cover the boundary, ∂S . Hence:

$$\partial S \subset \bigcup_{i=1}^m S_i \quad (2.3)$$

Hence, we can obtain an upper bound for the Lebesgue measure for the set ∂S from:

$$\begin{aligned}
\lambda_2(\partial S) &\leq \sum_{i=1}^m \lambda_2(S_i) \\
&\leq \sum_{m=1}^m \frac{16\pi^2}{m^2} \\
&\leq \frac{16\pi^2}{m}
\end{aligned} \tag{2.4}$$

Given any $\epsilon > 0$ we can find an integer $M \in \mathbb{N}$ such that $M > \frac{16\pi^2}{\epsilon}$ for which (2.4) is smaller than ϵ , whenever $m > M$, and we can conclude that, in $\mathbb{R}^2$, the set ∂S has measure zero.

Exercise 5 (Jordan null sets)

Let $A \subset \mathbb{R}^n$. We say that A is a *Jordan null set* if for any $\varepsilon > 0$ there exist finitely many (n -dimensional) open intervals $I_1, \dots, I_m$ with the property

$$\sum_{k=1}^m \lambda_n(I_k) < \varepsilon \quad \text{and} \quad A \subset \bigcup_{k=1}^m I_k.$$

Observe that by definition every Jordan null set is also Lebesgue null.

- (a) Show that any Jordan null set is necessarily bounded.
- (b) Give an example of an uncountable Jordan null set in $\mathbb{R}^2$.
- (c) Give an example of a Lebesgue null set in $\mathbb{R}$ that is *not* a Jordan null set.

Answer:

(a) Each interval in $(I_k)_{k=1}^m$ is bounded and this means that their (finite) union must also be bounded.

Indeed, let $I_k = (a_{k,1}, b_{k,1}) \times (a_{k,2}, b_{k,2}) \times \dots \times (a_{k,n}, b_{k,n})$, then it is clear that in each of the n dimensions we can create: $a_i = \min_k(a_{k,i})$ and $b_i = \max_k(a_{k,i})$. Then it is clear that:

$$\begin{aligned} I_k &\subset (a_1, b_1) \times (a_2, b_2) \times \dots \times (a_n, b_n) \quad \forall k = 1, 2, \dots, m \\ \implies A &\subset \bigcup_{i=1}^m I_k \subset (a_1, b_1) \times (a_2, b_2) \times \dots \times (a_n, b_n) \end{aligned}$$

□

(b) Consider the set: $S = (0, 1) \times \{0\}$ - it is definitely uncountable. Given $\epsilon > 0$, we can then consider the single open covering:

$$I_1 = (0, 1) \times (-\epsilon/3, \epsilon/3)$$

Clearly $S \subset I_1$ and $\lambda_2(I_1) = (1 - 0) \frac{2}{3} \epsilon < \epsilon$. Hence, the set S is a Jordan null set.

(c) As Jordan null sets are always bounded we can easily find a Lebesgue null set that is not a Jordan null set.

Consider the set $\mathbb{N}$ - which is an **unbounded** countable collection of points - i.e. a countable collection of null sets and hence is a Lebesgue null set, but NOT a Jordan null set.